Tutorial - 9

Customer ID	Credit Score	Annual Income	loan Amount	Employment years	Default
1	650	45	150	2	1
2	750	80	200	8	0
3	620	38	180	1	1
4	780	95	250	10	0
5	640	42	160	2	1
6	760	85	220	9	0
7	630	40	170	1	1
8	770	90	230	10	0
9	660	48	155	3	1
10	740	78	210	7	0
11	625	36	175	1	1
12	785	92	240	11	0
13	655	46	165	2	1
14	755	82	215	8	0
15	635	41	172	2	1
16	775	88	235	9	0
17	645	44	158	2	1
18	765	86	225	8	0
19	628	39	168	1	1
20	780	94	245	0	0

Step-1 : Divide dataset (K-fold cross validation)

Total = 20 samples

Default (Target variable) = 1 ⇒ [10, Samples]

Default (Target variable) = 0 ⇒ [10, samples]

Each fold = 4, samples ⇒ Each fold consist of 2(1)s, 2(0)s

Step-2 :

Fold (1) : customer ⇒ 1(1), 2(0), 3(1), 4(0)

Fold 2 : customer ⇒ 5(1), 6(0), 7(1), 8(0)

Fold 3 : customers ⇒ 9(1), 10(0), 11(1), 12(0)

Fold 4 : customers ⇒ 13(1), 14(0), 15(1), 16(0)

Fold 5 : customers ⇒ 17(1), 18(0), 19(1), 20(0)

Step-3 : Select Fold (1)

validation set (Test) = customers (1, 2, 3, 4)

Training set = customers (5 to 20)

↳ Remaining customers.

Step-4 : Logistic Regression Formula

$$P = \frac{1}{1+e^{-z}}$$

$$z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 --- + x_n$$

$$Z = 15 - 0.02(\text{credit}) - 0.05(\text{Income}) + 0.03(\text{loan}) - 0.5(\text{employment})$$

1) $= 15 - 0.02(650) - 0.05(45) + 0.03(150) - 0.5(2)$

$= 3.25$

2) $= 15 - 0.02(750) - 0.05(80) + 0.03(200) - 0.5(8)$

$0.119 = -2.$

3) $= 15 - 0.02(620) - 0.05(38) + 0.03(180) - 0.5(1)$ 0.05

0.996 = 5.6

4) $15 - 0.02(780) - 0.05(95) + 0.03(250) - 0.5(10)$

$= -2.85$

Step:4 - calculate for each validation sample

customer 1 : $z = 15 - 0.02(650) - 0.05(45) + 0.03(150) - 0.5(2)$

$= 3.25$

$$P = \frac{1}{1+e^{-2}} = \frac{1}{1+e^{-3.25}} \simeq 0.962$$

$P = 0.962 \geq 0.5 \Rightarrow$ Default, [predicted 1]

Results :-

	Actual	predicted	probability
customer 1	1	1	0.962
customer 2	0	0	0.119
cust 3	1	1	0.996
cust 4	0	0	0.05

confusion Matrix -

	Predicted 1	Predicted 0
Actual 1	2	0
Actual 0	0	2

Calculate Matrix -

$$\text{Accuracy} \Rightarrow \frac{\text{Correct predictions}}{\text{Total predictions}} = \frac{2+2}{4}$$

$= 100\%$

$$\text{Precision} \Rightarrow \frac{TP}{TP+FP} = \frac{2}{2+0} = 1$$

$$\text{Recall} \Rightarrow \frac{TP}{TP+FN} = \frac{2}{2+0} = 1$$

$$F_1 \text{ Score} = \frac{2 \times \text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}} = 1$$